

***pdsA*, a Gene Involved in the Production of Active Phosphodiesterase during Starvation of *Dictyostelium discoideum* Amoebae**

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Summary. Upon starvation, amoebae of the mutant strain HPX235 are unable to aggregate. Previous work has shown that this aggregateless character was associated with a nearly complete block in the production of the phosphodiesterase by these cells. Aggregation of the HPX235 amoebae can be induced with exogenous phosphodiesterase. In the present work, we show that both the aggregateless character and the block in phosphodiesterase production appear to result from the same recessive mutation, allocated to l.g.IV. Two other mutant strains displaying a comparable phenotype (HPX262 and HP594) were shown by complementation to belong to the same locus *pdsA*. Unlike wild type cells, the mutants of the locus *pdsA* cannot be induced to produce phosphodiesterase following treatment of the cells with exogenous cAMP, whether exogenous phosphodiesterase is present or not in the starvation buffer. It is concluded that *pdsA* is either the structural gene for the phosphodiesterase or a controlling element whose integrity is required for phosphodiesterase production. Mutations in *pdsA* share secondary effects among which the abnormally low production of the phosphodiesterase inhibitor. However, this effect can be overcome upon addition of exogenous phosphodiesterase, and most likely results from the lack of cAMP hydrolysis.

The late development is also affected in *pdsA* mutants. Aggregates formed in the presence of exogenous phosphodiesterase cannot culminate normally. This suggests that the level of cAMP hydrolysis also plays a role during the late stages of development of *Dictyostelium discoideum*.

Introduction

The developmental program of *Dictyostelium discoideum* amoebae takes place under condition of food deprivation. The first part of this cycle is marked by the aggregation of the cell population. Starved cells develop within a few hours the ability to emit and respond to pulses of a chemotactic mediator, which was identified as 3',5'-cyclic AMP (cAMP). (Konijn, 1972). Pulses of cyclic AMP direct the chemotactic migration of the cell population toward centers where multicellular aggregates are formed. In a second step aggregates differentiate further into fruiting bodies which consist of a spore mass supported by a stalk.

During the aggregation phase, cyclic AMP pulses not only direct the chemotactic motion of the cells, but also induce the expression of some specialized functions which characterize "aggregation-competent" cells. (Darmon et al., 1975; Gerisch et al., 1975). The physical elimination of cAMP is thought to be of importance in preventing the signals from being masked by a "noise" of over-accumulated nucleotide (Shaffer, 1962). The degradation of the chemotactic mediator is effected by the cAMP phosphodiesterase. This enzymatic activity is developmentally regulated. It is induced by a high exogenous concentration of cAMP, or by low concentration pulses of this nucleotide (Klein, 1975; Klein and Darmon, 1975; Gerisch et al., 1975). The enzyme is found in the cytosol, associated with the membranes or is excreted in the medium (Malchow et al., 1972; Chang, 1968). While the two cell-associated activities reach maximal levels at the time of aggregation, the extracellular enzyme may be inactivated by a specific inhibitor which is also secreted into the extracellular medium (Riedel et al., 1971). Inhibitor production,

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Abbreviations used cAMP, adenosine 3',5'-cyclic monophosphoric acid; l.g.: linkage group; PDE: 3',5'-cAMP phosphodiesterase

unlike that of the phosphodiesterase is, depressed by cAMP (Klein and Darmon, 1977; Tsang and Coukell, 1977). In addition to the time course of phosphodiesterase production, the physiological importance of this enzyme has been established on several grounds (Alcantara and Bazill, 1976; Darmon et al., 1978). In contrast the mechanisms which control its expression at the genetic level remain poorly understood. The study of this problem was undertaken in the present work using the mutant strain HPX235 which presents a nearly complete deficiency of phosphodiesterase production during starvation (Barra, 1977; Darmon et al., 1978).

Material and Methods

Strains of *Dictyostelium discoideum*. The characteristics of the haploid strains used are detailed in Table 1. HPX200 is a spontaneous methanol-resistant mutant of X2. The parental strains TS12 and X2 were described by Katz and Sussman (1972) and Williams et al. (1974) respectively. Strains HPX235 and HPX262 were isolated as aggregateless derivatives of strain HPX200, while HP594 is an aggregateless mutant of strain TS12.

Culture and Starvation Conditions. Strains were grown on SM agar in association with *Enterobacter aerogenes* (Sussman, 1966). Growth phase cells were harvested from almost confluent SM agar plates. Amoebae were washed several times by centrifugation in 17 mM phosphate buffer, pH 6.2. In experiments dealing with enzymatic assays, amoebae were starved in shaken suspensions. Starvation was initiated by resuspending the washed cells in phosphate buffer at a final density of 10^7 amoebae/ml. Under such conditions, wild type amoebae differentiate into aggregation-competent cells within 7–10 h. To monitor microscopically the ability of the cells to aggregate, sample of $3-5 \times 10^6$ amoebae in 5 ml of buffer were poured in a 5 cm diameter Corning tissue culture dish (Lee, 1972). However, the last stage of the developmental program, the differentiation of the aggregates into fruiting bodies does not occur in a liquid phase. For the observation of this

stage, 10^7 cells were seeded on a Millipore membrane resting on a lower pad of filter paper soaked with 1 ml of buffer (Sussman, 1966).

When the HPX235 cells were starved in the presence of exogenous phosphodiesterase, the supernatant buffer in which the mutant strain HP550 had been starved for 24 h was used as a crude source of phosphodiesterase. HP550 amoebae accumulate the phosphodiesterase during starvation but do not produce the inhibitor (Barra, unpublished). Aliquots resulting in a final activity of 10 units/ml (defined below) were added to the buffer at the beginning of the starvation. To this supernatant could be substituted purified beef-heart PDE at a final concentration of 5 units/ml.

Treatment with cAMP. Cell suspensions in 17 mM phosphate buffer were either made in 10^{-3} M cAMP, or treated with externally imposed cAMP pulses applied at 5 min intervals. Each pulse resulted in an input of 10^{-7} M cAMP.

Assay of Phosphodiesterase and Inhibitor. After various periods of time during starvation, phosphodiesterase and inhibitor activities were assayed by the methods of Brooker et al. (1968) and Riedel et al. (1972) respectively. Extracellular phosphodiesterase activity and inhibitor activity were measured in the supernatant of the starved cells. Cellular phosphodiesterase activity, which includes both the activity of the cytoplasmic and membrane species, was measured in the lysate obtained by freezing the cell pellet. One phosphodiesterase unit is defined as the amount of enzyme required to hydrolyse 1 nmole of cAMP per min at 30°. One inhibitor unit is defined as the amount of inhibitor necessary to inactivate 1 unit of phosphodiesterase activity from strain HP550.

Genetic Analysis. The general methods for the analysis were described by Williams et al. (1974).

a) Isolation of Diploid Strains. A loop of cells of two haploid strains, each carrying a recessive, non-allelic mutation thermosensitive for growth (*gts*) was suspended in 0.2 ml of phosphate buffer. The cell mixture was shaken in a vial for 20–24 h at 22° C, at 150 cycles $\cdot$ min $^{-1}$. The content of each vial was spread in association with bacteria on SM agar plates. Plates were incubated at the restrictive temperature (27°) for five days. Temperature-resistant plaques appeared at low frequency (ca. 10^{-6}). Amoebae of several plaques were purified by three clonal isolations at high temperature before being used in a segregation analysis. Each haploid strain

Table 1. Description of the haploid strains of *Dictyostelium discoideum* used

Strain	Origin	Genotype ^a				Phenotype ^b					
		linkage groups				Cy	Meth	G. Res	Sp.Col	Pig	Agg
		I	II	III	IV						
TS12	Katz	<i>cycA</i> ₁	<i>whi</i> , <i>tsgD12</i>			R	S	NG	W	NB	P
HPX200	this lab.		<i>acrA</i> ₁	<i>tsgA</i> ₁	<i>bwn</i>	S	R	NG	Y	B	P
HPX235	this lab.		<i>acrA</i> ₁	<i>tsgA</i> ₁	<i>bwn</i> , <i>pds A550</i>	S	R	NG	—	B	N
HPX262	this lab.		<i>acrA</i> ₁	<i>tsgA</i> ₁	<i>bwn</i> , <i>pdsA551</i>	S	R	NG	—	B	N
HP 594	this lab	<i>cycA</i> ₁	<i>whi</i> , <i>tsgD12</i>		<i>pds A552</i>	R	S	NG	—	NB	N

^a Only the markers used in this work are indicated. The symbols used to describe mutant loci were defined previously (Williams et al., 1974). *cycA*₁ allows growth in the presence of cycloheximide (300 µg/ml). *acrA*₁ allows growth in the presence of acridine or methanol (2%). *tsg* prevents growth at restrictive temperature (27°), *whi* confers white spore color, wild allele confers yellow spore color, *bwn* confers the ability to secrete brown pigment

^b The phenotypes corresponding to the different mutations are symbolised as follows: (Cy) cycloheximide; (Meth) methanol; (G. Res) growth at restrictive temperature (27°); (Sp Col.) Spore color; (Pig) pigment; (Agg) aggregation; (R) resistant; (S) sensitive; (W) white; (Y) yellow; (B) brown; (NB) not brown; (NG) no growth; (P) wild aggregation (N) aggregateless

carried several recessive markers (Table 1). Therefore the wild type phenotype for these markers, as well as the ability to yield segregant amoebae with either of these characteristics, was taken as a criterion for diploidy. When diploid amoebae were able to fructify, the large spore size characteristic of diploid cells was used as an additional criterion

b) Segregation Analysis. Spores or amoebae from diploid clones which were heterozygous for drug resistance markers were plated with *E. aerogenes* at $5 \cdot 10^4$ cells by plate on SM agar containing the appropriate inhibitor (cycloheximide 500 $\mu\text{g}/\text{ml}$ or 2% methanol). Resistant segregants from plates containing either of these inhibitors were next analyzed for the unselected markers: colour markers (white or yellow spores, brown pigment production), resistance to the other inhibitor, ability to grow at 27°C (Williams et al., 1974).

Chemicals. Phosphodiesterase 3',5'-cyclic nucleotide from beef heart, snake venom of *Ophiophagus hannah*, adenosine 3',5'-cyclic monophosphoric acid were obtained from Sigma, [^3H]adenosine 3',5'-cyclic monophosphate from CEA and anion exchange resin AG1-X10, 200-400 mesh, chloride form, from Bio-Rad.

Results

Nature of the Morphological Mutation of Strain HPX235

This strain is aggregateless. During starvation under submerged conditions on Petri dishes, HPX235 amoebae remain independent. The physiology of this strain has been described (Barra, 1977). Supplying the starvation buffer with phosphodiesterase stimulates HPX235 cells to aggregate as wild type parental HPX200 cells do. Assays of cellular and extracellular phosphodiesterase activity during development indicates a practically total deficiency in the production of phosphodiesterase by the HPX235 cells (Darmon et al., 1978). Various treatments with cAMP have been reported to stimulate the production of phosphodiesterase by wild cells during starvation (Klein 1975). Figure 1a shows that the phosphodiesterase activity of starved HPX200 wild cells appears three hours earlier when these cells are submitted to 10^{-7} M cAMP pulses; similarly, addition of 10^{-3} M cAMP in the buffer at the beginning of starvation results in a precocious production of cellular phosphodiesterase activity (6 h earlier than control) and causes the extracellular phosphodiesterase activity to reach levels 6 to 7 fold higher than that of control cells (Fig. 1b). None of these effects is observed when the HPX235 cells are submitted to the same treatments, the level of cell-associated enzyme remaining lower than 0.2 units per 10^7 cells (compare with Fig. 1a). Even in the presence of exogenous phosphodiesterase, i.e. under conditions which restore a wild type pattern of aggregation, neither of the treatments induces significant phosphodiesterase production in HPX235 amoebae.

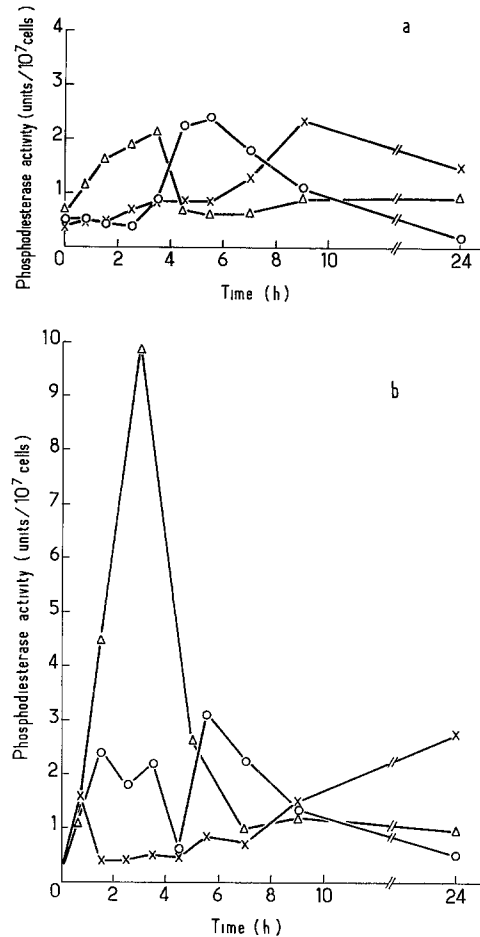


Fig. 1a and b. Effects of cAMP on production of phosphodiesterase by HPX200. Conditions of starvation: see text a Cellular activity and b extracellular activity. Control (x), cAMP pulses (o), single addition of 10^{-3} M cAMP (Δ)

As shown in Fig. 2, the phosphodiesterase inhibitor is produced by HPX200 as early as 3 hours after starvation. Addition of 10^{-3} M cAMP at the beginning of starvation delays the appearance of this activity by about 3 h but does not interfere with the final burst. Pulses of cAMP significantly reduce inhibitor production. Secretion of inhibitor by starved HPX235 mutant cells is very reduced as compared to HPX200 and is detected only after prolonged periods of starvation (Fig. 3). However when these cells are starved in the presence of exogenous phosphodiesterase (either purified beef heart PDE or crude supernatant of HP550), their secretion of phosphodiesterase inhibitor becomes identical to that of wild type cells (Fig. 3). Inhibitor activity then responds to treatments by cAMP in a way comparable to that of HPX200 cells (Fig. 2). These results suggest that one (or several) mutation(s) in the strain HPX235 affect(s) primarily the production of phosphodiesterase and that the abnormality of inhibitor production is a second-

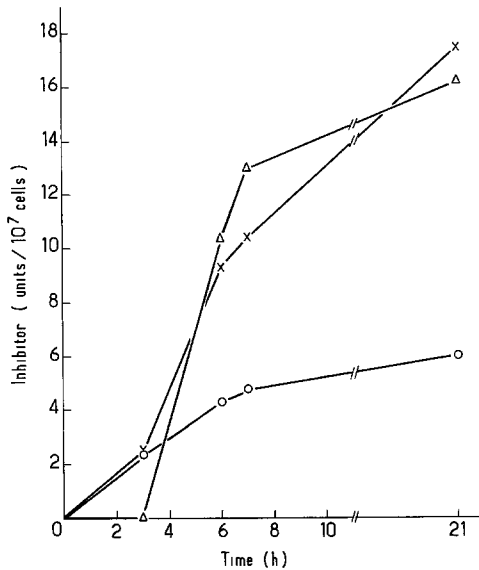


Fig. 2. Phosphodiesterase inhibitor produced by HPX200 cells during starvation in phosphate buffer. Inhibitor activity in the starvation medium was determined as described in Material and Methods. Same conditions of starvation as for Fig. 1a and b: Control (x), cAMP pulses (o), single addition of 10^{-3} M cAMP (Δ)

any effect of the deficiency in phosphodiesterase activity. The mutation(s) interfering with phosphodiesterase production will hereafter be referred to as *pds*.

Location of the *pds* Mutation

The parasexual genetic system of *D. discoideum* can be used to determine the location of developmental mutations in established linkage groups (Coukell, 1975; Williams and Newell, 1976). Diploids are formed by fusing HPX235 cells with an *Agg*⁺ haploid strain, TS12. Both strains carry recessive markers (Material and Methods and Table 1) that allow us to select segregant amoebae for linkage group I (*cycA*₁) or linkage group II (*acrA*₁) and to identify linkage group III and IV (*tsgA*₁ and *bwn*). Diploids have a wild type aggregation phenotype, a yellow spore color and a sensitivity to cycloheximide and methanol drugs phenotype.

During the analysis of haploid segregants, the *pds* mutation was first identified on the basis of the *Agg*⁻ phenotype of the haploid amoebae (Brachet et al., 1977). An analysis of segregants selected on methanol-containing agar showed that the *pds* mutation conferring the *Agg*⁻ phenotype is linked to *bwn* and must therefore be located in linkage group IV (Table 2). However, all the *Agg*⁺ clones showed abnormal fructification. This is most easily interpreted by assuming that a second mutation linked to the *acrA*₁ marker inter-

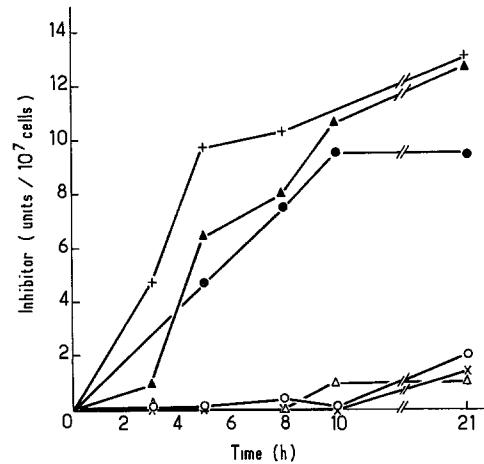


Fig. 3. Phosphodiesterase inhibitor produced by HPX235 mutant cells. The inhibitor activity produced by the amoebae starved in phosphate buffer untreated (x), or subjected to continuous pulses of cAMP (o) or to a single addition of 10^{-3} M cAMP at the time of starvation (Δ) is compared with the activity produced by amoebae starved in the same conditions but in the presence of 8 units/ml of exogenous PDE: control (+), cAMP pulses (●) or single initial addition of 10^{-3} M cAMP (▲)

Table 2. Assignment of the *pds* mutation to linkage group IV by haploidisation of three independent diploids HPX235/TS12 (DP1, DP5, DP7)

Marker selected	Non selected markers (linkage groups)							
	cycA ₁ tsgA ₁		pds		clone number			
	(I)	(III)	(IV)	(?)	Dp1	Dp5	Dp7	total
acrA ₁ (II)	A ₁	A ₁	bwn	pds	5	4	12	21
	A ₁	A ₁	+	+	8	6	7	21
	A ₁	+	bwn	pds	5	5	13	23
	A ₁	+	+	+	4	6	12	22
	+	A ₁	bwn	pds	2	1	3	6
	+	A ₁	+	+	4	6	1	11
	+	+	bwn	pds	2	3	2	7
	+	+	+	+	6	4	2	12

Genotype are inferred from the following phenotypes: cycloheximide resistant clones (*cycA*₁), growth at restrictive temperature (*tsgA*₁), secretion of brown pigment (*bwn*), no aggregation (*pds*)

feres with late development. Further support for these interpretations was obtained from an analysis of segregants selected on cycloheximide. In this case, a normal fructification phenotype was found among the *Agg*⁺ clones carrying the *acrA*₁⁺ marker (linkage group II from TS12), and an abnormal phenotype among those carrying the *acrA*₁ marker (l.g.II from HPX235). The mutation causing the *Agg*⁻ phenotype

Table 3. Aggregation and phosphodiesterase production during starvation of haploid segregant cells of HPX235/TS12 diploids

Genotype	Number of clone	Aggregation phenotype		Phosphodiesterase activity during starvation
		without exogenous PDE	with exogenous PDE	
<i>acrA₁ pds bwn</i>	17 (4)	—	+	deficient
<i>acrA₁⁺ pds bwn</i>	20 (6)	—	+	deficient
<i>acrA₁ pds⁺ bwn⁺</i>	6 (3)	+	+	wild type

— aggregateless (cells remained independent)

+ wild type aggregation

Cells were starved in Petri dishes as described in Material and Methods. Phosphodiesterase activity was assayed during starvation in cells of a few clones of each genotype studied (number in parentheses)

was linked to the *bwn* marker. Thus, in agreement with the analysis carried out on methanol plates, the *pds* mutation was assigned to linkage group IV.

In order to support the above mentioned assumption that the failure to aggregate and the deficiency in phosphodiesterase production are two phenotypes of the *pds* mutation, a physiological study of the aggregation of 37 *Agg*[−] and 6 *Agg*⁺ segregants (randomly selected among 124 obtained on methanol containing plates), as well as an assay of the phosphodiesterase activity released during starvation by 10 of these *Agg*[−] clones and 3 *Agg*⁺ clones, was carried out. The results (Table 3) demonstrate the correlation between the *Agg*[−] phenotype and the deficiency in the production of phosphodiesterase.

Recessivity of *pds* Mutation

Two haploid segregants both of which carry the *pds* mutation but not the *acrA₁* linked mutation were isolated. They differ by their *tsg* mutation. HP37 arose in the above experiments while HP10 was obtained from a back-cross performed between HP37 and HPX200. Both strains are *agg*[−] but their aggregation can be induced in the presence of phosphodiesterase like HPX235. The genotype of these strains is as follows:

HP 37: $\frac{+}{cycA_1} \quad \frac{+}{+} \quad \frac{+}{whi} \quad \frac{+}{tsgD12}$

HP10: $\frac{+}{+} \quad \frac{+}{acrA_1} \quad \frac{+}{+} \quad \frac{+}{+}$

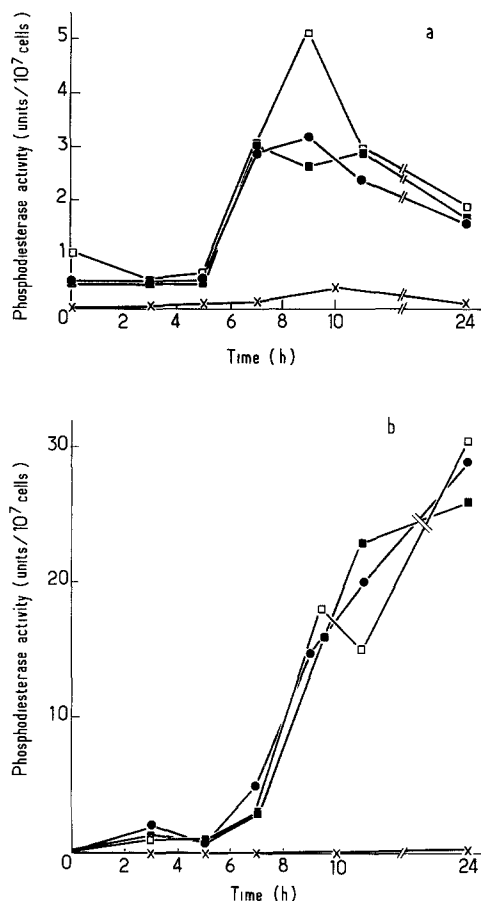


Fig. 4a and b. Phosphodiesterase produced by diploid cells heterozygous for the *pds* mutation. Same conditions of starvation as Fig. 1. **a** Cellular phosphodiesterase. **b** extracellular phosphodiesterase. Phosphodiesterase activity was determined in the parental haploid strains HP10 (x) and TS12 (●) and in two different diploids HP10/TS12 (□, ■)

Strain HP10 is an HPX200 strain in which linkage group IV was replaced by that of HPX235. Diploid strains obtained by fusing HP10 cells with TS12 cells exhibit a wild type aggregation phenotype, indicating that the *pds* mutation is recessive. The recessivity of the *pds* mutation was also verified with respect to phosphodiesterase production during starvation as shown in Fig. 4.

Homozygous *pds/pds* Diploids. Such diploids were obtained by fusing HP37 cells with HP10 cells. These *pds/pds* homozygous diploids had the *Agg*[−] phenotype and

the aggregation of these cells was induced by exogenous phosphodiesterase added during starvation. The phosphodiesterase activity was the same as that of the haploid parents HP37 and HP10, or that of HPX235.

Complementation Group. Fifteen independent aggregateless mutants were isolated, following three N-methyl N'-nitrosoguanidine mutageneses, whose phenotype is corrected by addition of crude supernatant of wild type cells starved for 8h or supernatant of the mutant HP550. 13 mutants derive from the haploid strain HPX200 and 2 from the haploid strain TS12.

Complementation tests were performed with the fifteen new mutants and HP10 or HP37, depending on the *tsg* mutation carried by the new mutant studied. Two different non-complementing diploids were repeatedly obtained: HP10/HP594 and HP37/HP262. A complementation group, *pdsA*, is defined by the three following strains: HPX235 (*pdsA550*), HPX262 (*pdsA551*) and HP564 (*pdsA552*). Phosphodiesterase activity and phosphodiesterase inhibitor activity developed during starvation by cells of HPX262 and HP564 mutants are very low as in the case of HPX235. Likewise, treatments of these new mutants by cAMP during starvation have no effect on phosphodiesterase production. The *agg*⁻ mutations of HPX262 and of HP564 are located in linkage group IV: the *pdsA551* mutation co-segregates with the *bwn* marker, and the *pdsA552* mutation co-segregates with the wild allele (*bwn*⁺).

Implication of pdsA Mutation in Post-Aggregative Development

Cells of strain HPX235 starved at high cell density on Millipore filters form aggregates, but their development stops at the finger stage. This phenotype might be a consequence of the mutation located in linkage group II. However, the segregant amoebae HP10 and HP37, which carry only the *pdsA* mutation, share a comparable defect in morphogenesis, although the formation of some spores inside the aggregates is occasionally observed. The two other mutants belonging to this complementation group, HPX262 and HP594, also exhibit this phenotype.

Discussion

The defect in phosphodiesterase production of the aggregateless mutants HPX235, HPX262 and HPX564 is due to mutations which define a comple-

mentation group, the locus *pdsA*, linked to the *bwn* marker in the linkage group IV. These mutations are recessive. The gene *pdsA* is primarily involved in phosphodiesterase production. Physiological analysis of HPX235 has shown that this strain does not produce significant amounts of either of the phosphodiesterase activities which develop during starvation: the cell-associated (cytosolic or membrane-associated) enzyme and the extracellular enzyme are present only as trace amounts. Moreover none of these activities can be induced by treatments which increase phosphodiesterase production in the wild type strain: addition of 10⁻³ M cAMP at the beginning of starvation or application of low concentration pulses of nucleotide. The *pdsA* gene therefore might be a structural gene common to the different forms of phosphodiesterase found in aggregating *Dictyostelium discoideum* amoebae; alternatively, it could be a control gene, but if several structural genes for phosphodiesterase exist, the *pdsA* gene must exert a control on all of them.

Secondary effects of *pdsA* affect the development of several functions characteristic of aggregation-competent amoebae, such as cAMP binding sites and contact sites A. However, when exogenous phosphodiesterase stimulates aggregation, HPX235 develops these functions normally (Brachet et al., 1979). These secondary effects are likely the results of the accumulation of cAMP in the medium (Darmon et al., 1978). Over-accumulation of cAMP also accounts for the abnormal production of PDE inhibitor by HPX235 amoebae. During starvation, cells synthesize no phosphodiesterase inhibitor. Addition of exogenous phosphodiesterase which depresses the levels of extracellular cAMP, causes the amoebae to produce inhibitor according to the same pattern as the wild type parental strain HPX200. Moreover, this production may be delayed or depressed, as happens in HPX200, if amoebae are treated with 10⁻³ M cAMP or with low concentration pulses of nucleotide. Thus HPX235 amoebae appear normal with respect to the negative regulation by cAMP of inhibitor production described in other strain (Klein and Darmon, 1977; Tsang and Coukell, 1977). The *pdsA* locus therefore is not involved in the control of inhibitor synthesis directly, but indirectly, through the control of the level of cAMP hydrolysis.

When starved in aggregative conditions, the three mutants fail to fructify. The cells form masses, inside which a few amoebae differentiate into spores. This suggests that the product of the *pdsA* gene may be involved in the late phases of development. Other aggregateless mutants which fail to produce phosphodiesterase during starvation and whose aggregation is induced by addition of supernatant of starved

wild cells, as for HPX235, show a similar postaggregative phenotype (unpublished data). These observations suggest that control for the cAMP level by phosphodiesterase is probably necessary for normal morphogenesis.

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